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Research Statement

Introduction

My research lies at the intersection of stable homotopy theory and arithmetic, with a particular emphasis on equivariant phenomena. The central goal of my research program is to understand *Real algebraic* K*-theory* through its duality phenomena and descent behavior. Duality concerns structural relationships among its invariants, while descent asks how global information can be recovered from local data. In parallel, my work on *Real trace methods* develops computational tools for Real algebraic K-theory.

My dissertation established the K*-theoretic Tate–Poitou duality* for number fields at prime 2, providing the non-equivariant starting point in this program. I am now extending this duality picture to Grothendieck–Witt theory, the fixed-point side of Real algebraic K-theory, toward a C_2-equivariant duality. In ordinary algebraic K-theory, arithmetic duality descends to the homotopical level. For Real algebraic K-theory, a corresponding equivariant cohomology theory is not yet available. I therefore approach the problem from the other direction, using the expected equivariant duality to guide the construction of a Real descent theory.

Another part of my research concerns the algebra intrinsic to genuine equivariance. My work on *transfer systems* and $\mathcal{RO}(C_2)$-gradings studies equivariant algebraic structure that disappear after forgetting the group action. Understanding these structures is a foundation in equivariant algebra and they also arise in any genuine equivariant form of descent. In the longer term, I aim to construct equivariant forms of cohomology and descent and to determine the arithmetic duality they carry.

1 Background

A basic example of duality in topology is the relation between homology and cohomology. The *Universal Coefficient Theorem* makes this relationship precise by identifying each cohomology group with the linear dual of the corresponding homology group when the coefficients lie in a field K:

$$H^n(X;k) \cong \mathrm{Hom}(H_n(X;k),k).$$

When a space carries a group action, some of its properties depend on the particular action. Two actions can have the same underlying object after the symmetry is forgotten while having very different equivariant information. For instance, reflection and rotation by a half-turn give two C_2-actions on a circle with different fixed-point sets, even though both have the same underlying circle.

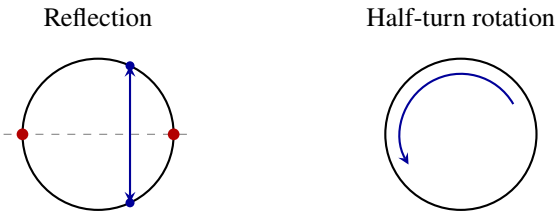


Figure 1: Reflection fixes two red points, while rotation by half a turn fixes none.

My work asks how duality and descent can be formulated without discarding the symmetry.

Real algebraic K-theory (KR) provides a natural setting for this question. Ordinary algebraic K-theory studies modules over rings, while KR-theory incorporates an involution and records the resulting C_2-equivariant structure. Forgetting this symmetry recovers K-theory, while taking fixed points leads to Grothendieck–Witt theory (GW). Thus, a duality in KR-theory should be visible both in K-theory and GW-theory, together with compatibility between them.

Arithmetic gives a concrete setting in which to study this interaction. In arithmetic, Tate–Poitou duality is expressed through étale cohomology, while Thomason descent connects this cohomological information to algebraic K-theory. Arithmetic duality should therefore also appear at the level of K-theory, and its Real refinement should preserve both the duality and the equivariant structure underlying descent.

2 Arithmetic Duality and Real Algebraic K-theory

Algebraic K-theory studies linear algebra over a ring R, packaging information about its general linear group $\mathrm{GL}(R)$ into topological invariants. Real algebraic K-theory brings ordinary algebraic K-theory and Grothendieck–Witt theory into a single C_2-equivariant framework in [HM13]. My dissertation established an arithmetic duality on the non-equivariant side through K-theoretic Tate–Poitou duality at the prime 2. My current work develops the corresponding fixed-point counterpart and shows that this duality extends across the hyperbolic sequence $\mathrm{U} \to \mathrm{K} \to \mathrm{GW}$. Together, these results suggest a larger equivariant duality in Real algebraic K-theory.

2.1 K-theoretic Tate–Poitou duality. Tate–Poitou duality is expressed through étale cohomology, while Thomason descent [Tho85] relates this cohomological information to $K(1)$-local algebraic K-theory. The corresponding K-theoretic problem is to realize Tate–Poitou duality at the homotopical level.

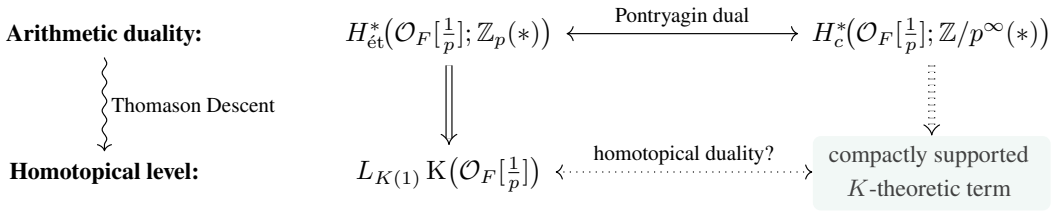
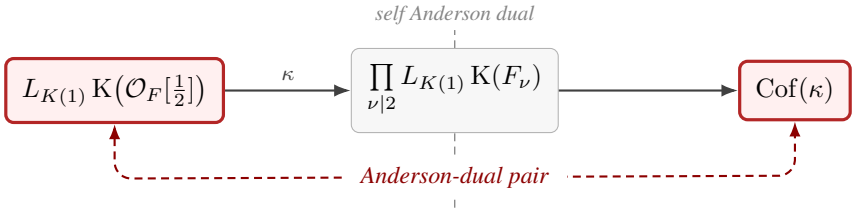


Figure 2: Cohomological duality suggests a homotopical duality after Thomason descent.

A homotopy-group-level form of this duality was conjectured by Calegari in [CE12]. Blumberg and Mandell established the homotopical-level local duality at every prime and the global duality at odd primes, both formulated in terms of p-adic Anderson duality [BM20]. The prime 2 remained open because the presence of real embeddings introduces additional cohomological behavior. My dissertation resolves this remaining case.

Theorem 1 ([Cho25])**.** *Let F be a number field and $\mathcal{O}_F[\frac{1}{2}]$ be its ring of 2-integers. The completion cofiber sequence of $K(1)$-local algebraic K-theory admits a canonical 2-adic Anderson self-duality. The duality acts by reflection, interchanging the end terms and identifying the middle term with its own dual.*



The theorem has a concrete application to trace methods in the algebraic K-theory of the sphere. For $F = \mathbb{Q}$, combining the duality theorem with known comparison results gives the following identification.

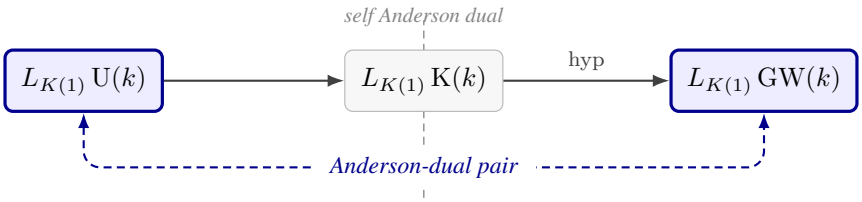
Corollary 2. *There is a canonical equivalence in nonnegative degrees*

$$I_{\mathbb{Z}_2} L_{K(1)}\,\mathrm{K}(\mathbb{Z}) \underset{[0,\infty)}{\simeq} \Sigma\,\mathrm{Fib}\Big(\mathrm{K}(\mathbb{S})^\wedge_2 \xrightarrow{\mathrm{trc}_{\mathbb{S}}} \mathrm{TC}(\mathbb{S})^\wedge_2\Big)\,.$$

Thus, arithmetic information in $K(\mathbb{Z})$ determines the fiber of the cyclotomic trace map for $K(\mathbb{S})$.

2.2 Grothendieck–Witt theory and duality. A fixed-point counterpart of the K-theoretic duality naturally leads to Grothendieck–Witt theory, which records modules equipped with nondegenerate bilinear forms [CDH+23]. The hyperbolic map $\mathrm{K} \to \mathrm{GW}$ relates algebraic K-theory to this additional form structure, whose homotopy fiber is known as U-theory. Their étale descent properties are well understood [BKSOsr15, BKOsr11]. The 2-primary descent techniques from my dissertation provide the arithmetic input for the GW-side. In current work with Lucy Yang, we show that over 2-adic local fields the resulting sequence has a duality symmetry.

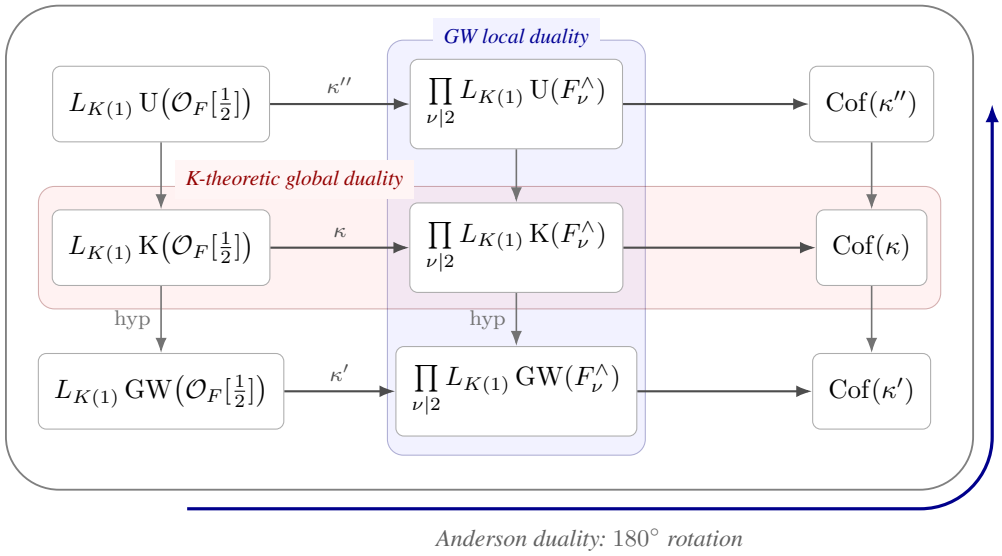
Theorem 3 ([CYon])**.** *Let k be a 2-adic local field. The hyperbolic sequence of $K(1)$-local* GW*-theory admits a canonical 2-adic Anderson self-duality. The duality acts by reflection, interchanging the end terms and identifying the middle term with its own dual.*



The significance of this theorem is that the duality is compatible across the hyperbolic sequence $\mathrm{U} \to \mathrm{K} \to \mathrm{GW}$ rather than appearing only in one theory at a time. It therefore suggests that the global duality should occur as one part of a larger self-dual arithmetic structure.

Our ongoing work seeks to globalize this duality to number fields.

Work in progress. *The completion sequences for* U*-theory,* K*-theory, and* GW*-theory fit into a (3×3) diagram compatible with the hyperbolic maps and local completion maps. The duality acts by half-turn rotation.*



Proving the self-duality of the full diagram would establish a global duality theorem for Grothendieck–Witt theory and combine the two results above in a single arithmetic structure. The rotational symmetry of the diagram leads to the following conjecture on equivariant duality in Real algebraic K-theory.

Conjecture. *Let F be a number field with trivial involution, and let*

$$\kappa_{\mathrm{KR}}^{[-1]} \colon L_{K(1)}\,\mathrm{KR}^{[-1]}\left(\mathcal{O}_F[\tfrac{1}{2}]\right) \to \prod_{\nu|2} L_{K(1)}\,\mathrm{KR}^{[-1]}\left(F_\nu^\wedge\right)$$

denote the completion map, where the superscript $[-1]$ *refers to the shift of the duality on perfect modules. There is a canonical equivalence of genuine* C_2*-spectra*

$$I_{\mathbb{Z}_2}^{C_2} L_{K(1)} \mathrm{KR}(\mathcal{O}_F[\tfrac{1}{2}]) \simeq \mathrm{Cof}\left(\kappa_{\mathrm{KR}}^{[-1]}\right)$$

where $I_{\mathbb{Z}_2}^{C_2}$ *denotes genuine* C_2*-equivariant* 2*-adic Anderson duality.*

After forgetting the C_2-action, the shift of the duality disappears, and the conjecture recovers the K-theoretic Tate–Poitou duality of my dissertation.

The conjecture specifies the duality that a Real descent theory should produce. Such a theory should recover ordinary descent after forgetting the C_2-action and be compatible with the expected global Grothendieck–Witt duality in the 3×3 diagram. These requirements guide my search for the equivariant cohomology theory underlying Real descent.

3 Real trace methods

Trace methods have long provided such tools in ordinary algebraic K-theory by approximating K-theory with more computable invariants such as topological Hochschild homology (THH) [DGM13]. These invariants also enter arithmetic duality. In [Cla17], Clausen established a cofiber sequence

$$\mathrm{TC}(\mathbb{Z}_p)^\wedge_p \longrightarrow L_{K(1)}\,\mathrm{K}(\mathbb{Q}_p) \xrightarrow{\simeq} I_{\mathbb{Z}_p}\big(L_{K(1)}\,\mathrm{K}(\mathbb{Q}_p)\big) \longrightarrow I_{\mathbb{Z}_p}\big(\mathrm{TC}(\mathbb{Z}_p)^\wedge_p\big)\ ,$$

connecting topological cyclic homology (TC) to the local duality appearing in Theorem 1.

A Real analogue of this sequence would connect Real topological cyclic homology (TCR) to equivariant duality in KR. This motivates developing Real trace methods to study both computations and duality in KR. In joint work with Teena Gerhardt, Liam Keenan, Juan C. Moreno, and J.D. Quigley, I develop *homological Real trace methods* that incorporate the additional equivariant structure present in the Real setting.

Ordinary THH carries a circle action, while Real topological Hochschild homology (THR) extends this to an $O(2)$-action by incorporating a reflection symmetry. We use this equivariant structure to compute



Figure 3: The $\mathbb{T}$-action on THH extends in THR to an $O(2) \simeq \mathbb{T} \rtimes C_2$-action.

the continuous homology of TCR^- [QS22]. Our method is based on a multiplicative spectral sequence compatible with equivariant power operations.

Theorem 4 ([CGK+26])**.** *For a* C_2*-equivariant commutative ring* A*, there is a natural multiplicative spectral sequence*

$$H_\star\big(\mathrm{THR}(A); \underline{\mathbb{F}}_2\big)\langle \bar{y} \rangle \Longrightarrow H^c_\star\big(\mathrm{TCR}^-(A); \underline{\mathbb{F}}_2\big),$$

compatible with equivariant power operations. For $MU_{\mathbb{R}}$*,* $BP_{\mathbb{R}}$*, and several truncated Brown–Peterson theories, we determine the differentials and the* E_∞*-page of the resulting spectral sequences.*

Our construction is a Real refinement of the homological trace methods of Bruner and Rognes [BR05], which played an important role in the study of Segal conjectures for the THH of bordism theories, MU and BP in [LeNR11]. A natural next test for our methods is a Real Segal conjecture for $MU_{\mathbb{R}}$ and $BP_{\mathbb{R}}$. Proving

this conjecture would advance computations of Real trace invariants for these theories and provide input for investigating a Real counterpart of the K–TC duality described above.

These computations rely on representation gradings and genuine equivariant operations. Understanding how these structures fit together is also necessary for an equivariant theory of descent.

4 Equivariant algebra: operations and gradings

An equivariant theory of descent must account for algebraic structure that disappears after forgetting the group action. Two features are particularly important: the equivariant operations carried by the coefficients and the representation gradings on which these operations act. My work investigates these two aspects of equivariant algebra through transfer systems and representation-graded Mackey and Tambara functors. These projects also clarify the kind of coefficient structure that a future equivariant descent theory should accommodate.

4.1 Equivariant operations. In equivariant stable homotopy theory, additive and multiplicative structures take forms that do not occur in ordinary algebra. Transfer maps play the role of equivariant addition, while norm maps play the role of equivariant multiplication. Each of these operations is parametrized by an N_∞-operad, and a pairing of the two operads encodes an equivariant distributive law [BH22]. *Transfer systems* describe N_∞-operads combinatorially, and the distributive law becomes a compatibility condition between two such systems [Rub21]. The realization problem asks whether every compatible pair of transfer systems arises from a pairing of N_∞-operads.

In joint work with David Chan, David Mehrle, Pablo Sanchez Ocal, Angélica Osorno, Ben Szczesny, and Paula Verdugo, we study the realization problem relating these two descriptions.

$$\begin{array}{ccc} \text{pairing of} & \Longrightarrow & \text{compatible pair of} \\ N_\infty\text{-operads} & \overset{\text{partially}}{\dashleftarrow} & \text{transfer systems} \end{array}$$

Theorem 5 ([CCM+25])**.** *A pairing of N_∞-operads determines a compatible pair of transfer systems. Conversely, if the additive transfer system is complete, then the compatible pair can be realized by a pairing of N_∞-operads.*

Beyond the complete case, we establish several additional families of realizable pairs and conjecture that every compatible pair of transfer systems is realizable.

4.2 Representation gradings. Equivariant descent spectral sequences are built from layers whose homotopy groups carry representation-graded Mackey functor structures. To use algebraically defined coefficients in such constructions, one must understand when they arise from equivariant spectra such as Eilenberg–Mac Lane or slice spectra. This leads to a realization problem for representation-graded Mackey functors.

For a C_2-spectrum X, its stable homotopy groups $\underline{\pi}_\star X$ form Mackey functors graded by representations ring $RO(C_2)$. An ordinary $RO(C_2)$-grading records only the class of a representation and not its symmetries. The category $\mathcal{RO}(C_2)$ introduced by Angeltveit and Bohmann in [AB18] refines this grading by also recording automorphisms of individual representation degrees.

In ongoing work with Noah Wisdom, we study the algebraic structure forced by these representation gradings. Motivated by the structure of stable homotopy groups, we call an $\mathcal{RO}(C_2)$-graded Mackey or Tambara functor *topological* when its grading data factors through its groupoid completion. For C_2, we show that the additional automorphisms recorded by a topological $\mathcal{RO}(C_2)$-grading are already encoded by the



Figure 4: $\mathcal{RO}(C_2)$-grading records the symmetries of the representation degree V.

canonical Burnside action on a Mackey functor. Thus the refined grading introduces no additional realization condition at this level.

This separates the symmetry of the grading from the remaining realization problem. Given a topological $RO(C_2)$-graded Mackey functor $M_\star$, when does there exist a genuine G-spectrum X such that

$$M_\star \cong \underline{\pi}_\star X?$$

Understanding this realization problem is a step toward determining which $\mathcal{RO}(C_2)$-graded coefficients can occur in equivariant descent for Real algebraic K-theory.

We also study concrete realized examples. In particular, we investigate the graded Tambara structure on the homotopy groups of equivariant complex bordism MU_G. Since MU_G is already a genuine G-spectrum, it provides a test case for understanding how representation gradings, transfers, and norm maps interact in realizable graded Tambara functors.

5 Outlook

My long-term goal is to understand how arithmetic duality extends to Real algebraic K-theory of rings with involution. The number-field results provide concrete models of duality, while Real trace methods offer a route toward investigating similar structures beyond the arithmetic setting. I seek an equivariant cohomological framework that explains these dualities and supports descent to KR, with coefficients that incorporate representation gradings and genuine equivariant operations.

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